

3.3.13 Amino acids, proteins and DNA (A-level only)

Amino acids, proteins and DNA are the molecules of life. In this section, the structure and bonding in these molecules and the way they interact is studied. Drug action is also considered.

3.3.13.1 Amino acids (A-level only)

Content	Opportunities for skills development
Amino acids have both acidic and basic properties, including the formation of zwitterions. Students should be able to draw the structures of amino acids as zwitterions and the ions formed from amino acids: • in acid solution • in alkaline solution.	

3.3.13.2 Proteins (A-level only)

Content	Opportunities for skills development
Proteins are sequences of amino acids joined by peptide links. The importance of hydrogen bonding and sulfur–sulfur bonds in proteins. The primary, secondary (α-helix and β-pleated sheets) and tertiary structure of proteins. Hydrolysis of the peptide link produces the constituent amino acids. Amino acids can be separated and identified by thin-layer chromatography. Amino acids can be located on a chromatogram using developing agents such as ninhydrin or ultraviolet light and identified by their R_f values. Students should be able to: • draw the structure of a peptide formed from up to three amino acids • draw the structure of the amino acids formed by hydrolysis of a peptide • identify primary, secondary and tertiary structures in diagrams • explain how these structures are maintained by hydrogen bonding and S–S bonds • calculate R_f values from a chromatogram.	

3.3.13.3 Enzymes (A-level only)

Content	Opportunities for skills development
Enzymes are proteins. The action of enzymes as catalysts, including the concept of a stereospecific active site that binds to a substrate molecule. The principle of a drug acting as an enzyme inhibitor by blocking the active site. Computers can be used to help design such drugs. Students should be able to: explain why a stereospecific active site can only bond to one enantiomeric form of a substrate or drug.	

3.3.13.4 DNA (A-level only)

Content	Opportunities for skills development
The structures of the phosphate ion, 2-deoxyribose (a pentose sugar) and the four bases adenine, cytosine, guanine and thymine are given in the Chemistry Data Booklet. A nucleotide is made up from a phosphate ion bonded to 2-deoxyribose which is in turn bonded to one of the four bases adenine, cytosine, guanine and thymine. A single strand of DNA (deoxyribonucleic acid) is a polymer of nucleotides linked by covalent bonds between the phosphate group of one nucleotide and the 2-deoxyribose of another nucleotide. This results in a sugar-phosphate-sugar-phosphate polymer chain with bases attached to the sugars in the chain. DNA exists as two complementary strands arranged in the form of a double helix. Students should be able to: explain how hydrogen bonding between base pairs leads to the two complementary strands of DNA.	

3.3.13.5 Action of anticancer drugs (A-level only)

Content	Opportunities for skills development
The Pt(II) complex cisplatin is used as an anticancer drug. Cisplatin prevents DNA replication in cancer cells by a ligand replacement reaction with DNA in which a bond is formed between platinum and a nitrogen atom on guanine. Appreciate that society needs to assess the balance between the benefits and the adverse effects of drugs, such as the anticancer drug cisplatin. Students should be able to: • explain why cisplatin prevents DNA replication • explain why such drugs can have adverse effects.	